MILITARY SPECIFICATION

PROJECTOR, RETICLE IMAGE: 11744800 M21

This specification is approved for use by the US Army Armament Research and Development Command (ARRADCOM) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements and quality assurance provisions for the Projector, Reticle Image: 11744800 M21 which is a unit of the M21 Ballistic Computer System, 11732979.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-N-411 Nitrogen, Technical

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-STD-45662	Calibration System Requirements

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-810	Environmental Test Methods
MIL-STD-1275	Electrical Circuit, 28 Volt D.C. Transient Characteristics for Military Vehicles

DRAWINGS

ARRADCOM

11744800	Projector, Reticle Image: M21

Inspection Equipment

11746741	Fixture, Collimation
11750233	Test Set - Computer FAT
11821260	Test Fixture - Electrical, Test Set
11821457	Fixture, Shock, Vertical and Horizontal Plane
11821458	Shock Adapter, Plate Vib Horz-Vert Plane/ Shock/Z
11821465	Unit Fixture, Reticle Projector Vib/Shock/Z
11821748	Test Set - Immersion Leak, Purge and Fill

Packaging Data Sheet

P11744800	Packaging of Projector, Reticle Image: M21

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication. The reticle image projector, herein referred to as the projector, shall be manufactured in accordance with Drawing 11744800 and drawings pertaining thereto and, when assembled, shall meet the requirements of this specification. (See 4.5.1)

3.1.1 Function. The projector shall provide the following functions:

The projector shall project and automatically position laterally, a reticle pattern into the periscope optical system for control of the main gun and project a reticle pattern into the periscope unity power viewer for control of the machine gun.

3.1.2 General specifications. The following provisions of MIL-F-13926 apply: (See 4.5.1)

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.1.3 Ambient conditions. Standard ambient conditions shall be as follows:

a.	Temperature	73° ± 18 °F
b.	Relative humidity	50 percent ± 30 percent
c.	Atmospheric pressure	28.5 + 2.0 -3.0 in. Hg.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample units for first article inspection and approval (see 4.4 and 6.2).

3.3 Performance. Unless otherwise specified, the assembly shall meet the performance requirements specified herein under the following conditions:

a. standard ambient conditions of 3.1.3.
b. boresight scale indexing of 3.3.1.

3.3.1 Boresight scale indexing. Each main gun system boresight knob scale shall be set to (4) with the following conditions established concurrently: (See 4.6.2.1)

a. The lower surface of the housing key horizontal within ± 15 minutes.

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b. The housing mounting surface adjacent to the key vertical within one degree.

c. Null setting of 0.0000 (electrical zero).

3.3.2 Machine gun system.

3.3.2.1 Sight. The line of sight of the center of the circular reticle shall subtend an angle of 26° ± 3 minutes with the lower surface of the key and shall be perpendicular to the vertical housing mounting surface within plus or minus three minutes. (See 4.6.2.2)

3.3.2.2 Focus. The focus of the reticle shall be not greater than -0.50 and not less than -1.0 diopter. (See 4.6.2.3)

3.3.2.3 Boresight knob travel. The reticle shall move not less than 10 mils in each deflection direction and not less than 10 mils in each elevation direction with like rotation of the corresponding boresight knob. (See 4.6.2.4)

3.3.2.4 Torque. The torque required to rotate each boresight knob shall be not greater than 5 inch-pounds and not greater than 10 inch-pounds at the high and low temperatures of 3.4.3. (See 4.6.2.5)

3.3.3 Main gun system.

3.3.3.1 Reticle position. With the deflection plane defined as the lower surface of the key, the reticle deflection line shall be parallel to the deflection plane within ± 1.0 degree. (See 4.6.2.6)

3.3.3.2 Parallax. Parallax of the sight shall be not greater than 0.05 mil at not less than 1200 meters. (See 4.6.2.7)

3.3.3.3 Boresight knob travel. The reticle shall move not less than 11.7 mils in raised elevation and not less than 17.6 mils in lowered elevation with like rotation of the elevation boresight knob. The reticle shall move not less than 14.6 mils in each deflection direction with like rotation of the deflection boresight knob. (See 4.6.2.8)

3.3.3.4 Reticle shift. The line of sight shall shift not greater than 0.1 mil perpendicular to the deflection and elevation reticles with each disengagement and each direction change of the corresponding boresight knob. (See 4.6.2.9)

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3.3.3.5 Motor driven line of sight. The line of sight in elevation shall deviate not greater than 0.20 mils within a spread not greater than 0.4 mils with the reticle driven from right to left. (See 4.6.2.10)

3.3.3.6 Boresight line of sight. The lines of sight in elevation and deflection shall deviate not greater than 0.5 mils within a spread not greater than 1.0 mils with each opposite boresight knob rotated from stop to stop. (See 4.6.2.11)

3.3.3.7 Backlash. Elevation and deflection boresight knob backlash shall be not greater than 0.15 mils. (See 4.6.2.12)

3.3.3.8 Motor tachometer. A signal of 115 ± 1% volts, 400 ± 1% Hz shall be applied to the fixed phase, pins J1-A and J1-B. (See 4.6.2.13)

3.3.3.8.1 Reticle motion. With a 48 ± 0.5 volts, 400 ± 1% Hz signal applied to the control phase, pins J1-C and J1-F, the reticle motion shall be as follows:

a. With the control phase voltage in phase with the fixed phase voltage, the reticle shall move to the right.

b. With the control phase voltage 180° out of phase with the fixed phase voltage, the reticle shall move to the left.

c. The reticle shall slew from right to left and left to right in not greater than 2.0 seconds for each direction and in not greater than 5.0 seconds at the high and low temperatures of 3.4.3.

d. With the reticle motion to the right, the output voltage between pins J1-H and J1-L shall be not less than 7.5 volts peak-to-peak and in phase with the fixed phase voltage.

3.3.3.8.2 Starting voltage. With an 8 volt rms, 400 ± 1% Hz signal applied to the control phase, pins J1-C and J1-F, the motor shall operate in each direction.

3.3.3.9 Reticle traversing accuracy. Each reticle line of sight reading for each null setting specified in table I shall be within the specified tolerance of the corresponding reticle position value. (See 4.6.2.14)

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TABLE I. Reticle traversing accuracy.

Item	Null Setting	Reticle Position (± mils)
1	0.74747	36.5 ± 0.15
2	0.71677	35.0 ± 0.15
3	0.61438	30.0 ± 0.15
4	0.51198	25.0 ± 0.15
5	0.40958	20.0 ± 0.10
6	0.30719	15.0 ± 0.10
7	0.20479	10.0 ± 0.10
8	0.10240	5.0 ± 0.10
9	0.04096	2.0 ± 0.10
10	0.02048	1.0 ± 0.10

3.3.3.10 Zero line of sight. The reticle line of sight at electrical zero shall be perpendicular to the housing mounting surface within 0.15 mils after movement to the right and left. (See 4.6.2.15)

3.3.3.11 Limit switches. Referenced from zero, travel of the reticle not less than 36.5 mils and not greater than 40 mils in each deflection direction shall actuate a limit switch. Deactuation of the limit switch shall permit travel of the reticle to less than 36.5 mils from each direction. Actuation of each switch shall electrically short pins J1-H and J1-L. (See 4.6.2.16)

3.3.3.12 Illumination. Both the machine gun system reticle pattern and the main gun system reticle pattern shall illuminate with application of 24 volts DC to pins J1-G and J1-Z. Both patterns shall be uniform and free from ragged edges and breaks. (See 4.6.2.17)

3.3.3.13 Boresight knob disengagement. The force required to disengage the deflection and elevation boresight knobs shall be not less than one pound and not greater than three pounds. (See 4.6.2.18)

3.3.3.14 Boresight knob rotation. The torque required to rotate the deflection and elevation boresight knobs shall be not greater than four inch-pounds and not less than one-half inch-pounds and not greater than five and one-half inch-pounds and not less than one-half inch-pounds at the high and low temperatures of 3.4.3. (See 4.6.2.19)

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3.3.3.15 Rheostat knob rotation. The torque required to rotate the rheostat knob shall be not greater than four inch-pounds and not greater than 16 inch-pounds at the high and low temperatures of 3.4.3. (See 4.6.2.20)

3.3.3.16 Cleanliness. The projector shall meet the assembled optical system cleanliness requirements specified in MIL-O-13830. (See 4.6.2.21)

3.3.3.17 Sealing. With an internal dry nitrogen pressure of 5.0 ± 0.1 pounds per square inch gauge (psig), the projector shall exhibit a pressure drop not greater than 0.2 psig for not less than 2 hours. Each movable control device on the projector shall be exercised three times to its extreme positions during the second hour of this time period. The projector shall be purged with dry nitrogen and have a final pressure of 0.5 ± 0.1 psig. Nitrogen shall be BB-N-411, type I, grade B, class 1 with a dewpoint not greater than -25°F. (See 4.6.4)

3.3.3.18 Transient voltages. The projector shall be capable of operating as specified herein after exposure to the transient voltages specified in MIL-STD-1275. (See 4.6.3.10)

3.4 Environmental.

3.4.1 Shock. The projector shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to three half sine wave shock pulses of 40 ± 4 gravity unit (g) for a duration of 18 ± 3 milliseconds (msec) applied in each direction along three mutually perpendicular axes. In addition, the projector shall be capable of withstanding three half sine wave shock pulses of 100.0 ± 10.0 g for a duration of 1.5 ± 0.2 msec applied in each direction along three mutually perpendicular axes. (See 4.6.3.1)

3.4.2 Vibration. The projector shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to the vibration profile of figure 1. Duration of exposure shall be not less than 80 minutes in each of three mutually perpendicular axes. (See 4.6.3.2)

3.4.3 Operating temperature. The projector shall be capable of operating as specified herein over the operating temperature range of -25°F to +125°F. (See 4.6.3.3, 4.6.3.4 and 4.6.3.5)

3.4.4 Storage temperature. The projector shall be capable of operating as specified herein after exposure to storage temperatures ranging from -65°F to 160°F. (See 4.6.3.3, 4.6.3.4 and 4.6.3.5)

3.4.5 Humidity. The projector shall operate as specified herein at up to 100 percent relative humidity. (See 4.6.3.6)

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3.4.6 Altitude. The projector shall operate as specified herein under the following conditions:

a. during exposure to an atomospheric pressure equivalent to 10.000 feet above sea level.

b. after exposure to an atmospheric pressure equivalent to 40,000 feet above sea level for not less than 1 hour. (See 4.6.3.7)

3.4.7 Dust. The projector shall operate as specified herein after exposure to moving dust for not less than 12 hours at velocities up to 1,750 feet per minute. (See 4.6.3.8)

3.4.8 Salt fog. The projector shall operate as specified herein after exposure to a 5 percent sodium chloride atomized spray for not less than 48 hours. (See 4.6.3.9)

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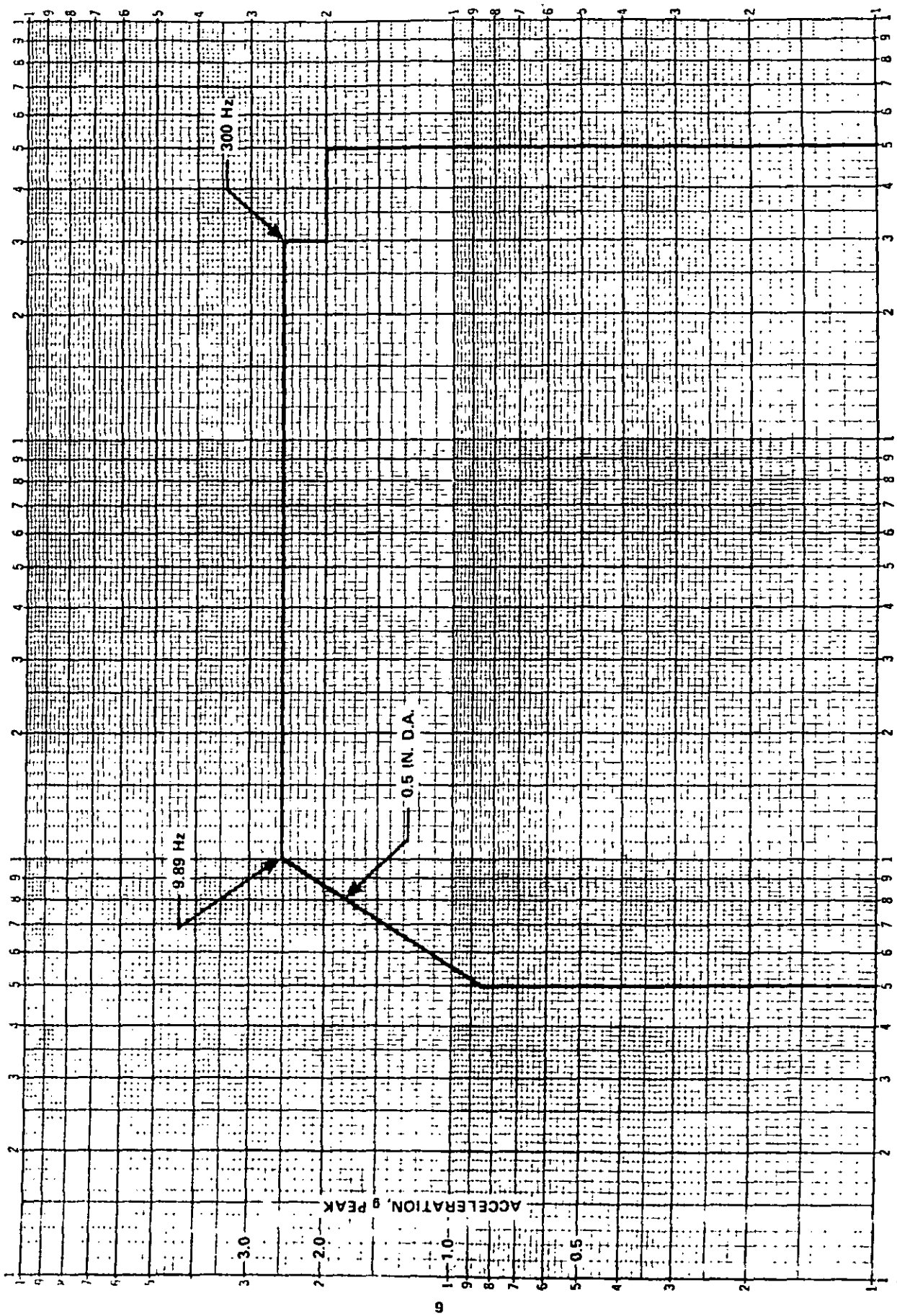


FIGURE 1. Vibration profile.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.4).
2. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-13926 and the conditions of 3.1.3.

4.4 First article. The requirements for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor shall be specified in the contract.

4.4.1 Sample. The first article sample shall be three projectors selected at random by the Government representative from the first five production projectors.

4.4.2 Inspection. The sample shall be subjected to all the inspections specified in tables II, III, IV and V.

4.4.3 Failure. Failure of any projector to meet any requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of any projector to comply with any stated requirement.

4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection with the sample and certificates of conformance for materials. The Government reserves the right to witness inspections performed by the contractor.

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TABLE II. First article tests.

Item	Characteristic	Requirement	Test Procedure
1	Humidity	3.4.5	4.6.3.6
2	Altitude	3.4.6	4.6.3.7
3	Dust	3.4.7	4.6.3.8
4	Salt Fog	3.4.8	4.6.3.9
5	Transient Voltage	3.3.3.18	4.6.3.10

4.5 Quality conformance inspection.

4.5.1 Procurement conditions. The following inspection plans shall apply where the projector is procured for use as a:

Use	Plan
a. Logistics spare projector apart from the next high assembly	A and C
b. Component assembly of the next higher assembly	B and C

4.5.2 Inspection plan A.

4.5.2.1 General sample and tests. One projector, as a control sample, shall be selected at random by the Government representative from each 100 projectors produced and shall be subjected to all the tests in table IV.

4.5.2.2 Environmental sample and tests. Three projectors, as a control sample, shall be selected at random by the Government representative from each 50 projectors produced or from each month's production, whichever occurs first. All the tests in table V shall be applied separately to each projector in the sample.

4.5.2.3 Acceptance. Where any one projector of either sample fails to meet the specified requirement, the lot shall be rejected. Rejected lots shall be subject to the provisions of MIL-STD-105.

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4.5.3 Inspection plan B.

4.5.3.1 Sample and tests. The sample shall be selected by the Government representative in accordance with the provisions of MIL-STD-105. All the tests in table IV and only the test of item 307 in table V shall be applied.

4.5.3.2 Acceptance. Acceptance and rejection shall be in accordance with MIL-STD-105.

4.5.4 Inspection plan C.

4.5.4.1 Sample and tests. Each projector in every lot shall be subjected to the tests in table III and shall be examined visually for completeness, improper assembly and evidence of poor workmanship.

4.5.4.2 Acceptance. Where any one projector fails to meet any specified requirement, the defective projector shall be removed from the lot and resubmitted only after all defects have been corrected.

4.5.5 Inspection equipment. Unless otherwise specified in the contract, the contractor shall supply, maintain and calibrate inspection equipment in accordance with the provisions of MIL-I-45607 and MIL-STD-45662.

4.5.6 Packaging inspection. The sampling and inspection of the preservation - packaging, packing and container marking shall be in accordance with the provisions of packaging data sheet P11744800.

TABLE III. Performance tests.

Item	Characteristic	Requirement	Test Procedure
101	Boresight scale indexing	3.3.1	4.6.2.1
102	Sight - machine gun system	3.3.2.1	4.6.2.2
103	Focus - machine gun system	3.3.2.2	4.6.2.3
104	Boresight knob travel - machine gun system	3.3.2.3	4.6.2.4
105	Torque - machine gun system	3.3.2.4	4.6.2.5

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TABLE III. Performance tests. - Continued

Item	Characteristic	Requirement	Test Procedure
106	Reticle position	3.3.3.1	4.6.2.6
107	Parallax	3.3.3.2	4.6.2.7
108	Boresight knob travel	3.3.3.3	4.6.2.8
109	Reticle shift	3.3.3.4	4.6.2.9
110	Motor driven line of sight	3.3.3.5	4.6.2.10
111	Boresight line of sight	3.3.3.6	4.6.2.11
112	Backlash	3.3.3.7	4.6.2.12
113	Motor tachometer	3.3.3.8	4.6.2.13
114	Reticle traversing accuracy	3.3.3.9	4.6.2.14
115	Zero line of sight	3.3.3.10	4.6.2.15
116	Limit switches	3.3.3.11	4.6.2.16
117	Illumination	3.3.3.12	4.6.2.17
118	Boresight knob disengagement	3.3.3.13	4.6.2.18
119	Boresight knob rotation	3.3.3.14	4.6.2.19
120	Rheostat knob rotation	3.3.3.15	4.6.2.20
121	Cleanliness	3.3.3.16	4.6.2.21
122	Sealing A	3.3.3.17	4.6.4

TABLE IV. General tests.

Item	Characteristic	Requirement	Test Procedure
301	Fabrication	3.1	Applicable Drawing - Visual
302	General specifications	3.1.2	MIL-F-13926 - Visual

TABLE V. Environmental tests.

Item	Characteristic	Requirement	Test Procedure
303	Shock	3.4.1	4.6.3.1
304	Vibration B	3.4.2	4.6.3.2
305	High Temperature	3.4.3 and 3.4.4	4.6.3.3
306	Low Temperature	3.4.3 and 3.4.4	4.6.3.4
307	Temperature Cycling	3.4.3 and 3.4.4	4.6.3.5
308	Sealing B	3.3.3.17	4.6.4

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4.6 Methods of inspection.

4.6.1 Test equipment. The test equipment in table VI shall be used where specified to perform the required test.

TABLE VI. Test equipment.

Item	Part Number	Nomenclature
1	11746741	Fixture, Collimation
2	11750233	Test Set - Computer, FAT
3	11821260	Test Fixture - Electrical, Test Set
4	11821457	Fixture, Shock, Vertical and Horizontal Plane
5	11821458	Shock Adapter, Plate Vib Horz-Vert Plane/ Shock Z
6	11821465	Unit Fixture, Reticle Projector Vib/Shock/Z
7	11821748	Test Set - Immersion Leak, Purge and Fill

4.6.2 Performance tests.

4.6.2.1 Boresight scale indexing. Use items 1 and 3 of table VI to determine compliance with 3.3.1.

4.6.2.2 Sight. Use items 1 and 3 of table VI to determine compliance with 3.3.2.1.

4.6.2.3 Focus. Use items 1 and 3 of table VI to determine compliance with 3.3.2.2.

4.6.2.4 Boresight knob travel. Use items 1 and 3 of table VI to determine compliance with 3.3.2.3.

4.6.2.5 Torque. Use items 1 and 3 of table VI to determine compliance with 3.3.2.4.

4.6.2.6 Reticle position. Use items 1 and 3 of table VI to determine compliance with 3.3.3.1.

4.6.2.7 Parallax. Use items 1 and 3 of table VI to determine compliance with 3.3.3.2.

4.6.2.8 Boresight knob travel. Use items 1 and 3 of table VI to determine compliance with 3.3.3.3.

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4.6.2.9 Reticle shift. Use items 1 and 3 of table VI to determine compliance with 3.3.3.4.

4.6.2.10 Motor driven line of sight. Use items 1 and 3 of table VI to determine compliance with 3.3.3.5.

4.6.2.11 Boresight line of sight. Use items 1 and 3 of table VI to determine compliance with 3.3.3.6.

4.6.2.12 Backlash. Use items 1 and 3 of table VI to determine compliance with 3.3.3.7.

4.6.2.13 Motor tachometer. Use items 1 and 3 of table VI to determine compliance with 3.3.3.8.

4.6.2.14 Reticle traversing accuracy. Use items 1 and 3 of table VI to determine compliance with 3.3.3.9.

4.6.2.15 Zero line of sight. Use items 1 and 3 of table VI to determine compliance with 3.3.3.10.

4.6.2.16 Limit switches. Use items 1 and 3 of table VI to determine compliance with 3.3.3.11.

4.6.2.17 Illumination. Use items 1 and 3 of table VI to determine compliance with 3.3.3.12.

4.6.2.18 Boresight knob disengagement. Use items 1 and 3 of table VI to determine compliance with 3.3.3.13.

4.6.2.19 Boresight knob rotation. Use items 1 and 3 of table VI to determine compliance with 3.3.3.14.

4.6.2.20 Rheostat knob rotation. Use items 1 and 3 of table VI to determine compliance with 3.3.3.15.

4.6.2.21 Cleanliness. The projector shall be examined in accordance with the assembled optical system cleanliness quality assurance provisions of MIL-O-13830.

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4.6.3 Environmental tests.

4.6.3.1 Shock.

4.6.3.1.1 Basic design. Use items 4 and 6 of table VI and subject the projector to the shock test specified in MIL-STD-810, method 516.2, procedure I, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 40 ± 4 g with a time duration of 18 ± 3 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the projector to the performance tests of 4.6.2.

4.6.3.1.2 High intensity. Use items 5 and 6 of table VI and subject the projector to the shock test specified in MIL-STD-810, method 516.2, procedure IV, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 100.0 ± 10.0 g with a time duration of 1.5 ± 0.2 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the projector to the performance tests of 4.6.2.

4.6.3.2 Vibration. Application of Vibration A and Vibration B shall be determined by the procurement conditions of 4.5.1 and the first article conditions of 4.4.

4.6.3.2.1 Vibration A. Use item 6 of table VI and subject the projector to the vibration test specified in MIL-STD-810, method 514.2, procedure VIII, except that the test level shall be the vibration curve specified in figure 1 and duration of exposure shall be not less than 80 minutes per axis. At the conclusion of this test, subject the projector to the performance tests of 4.6.2.

4.6.3.2.2 Vibration B. This test shall be conducted as in 4.6.3.2.1 except that the period of vibration shall be not less than 15 minutes in each axis.

4.6.3.3 High temperature. Subject the projector to the high temperature test specified in MIL-STD-810, method 501.1, procedure I. The highest operating temperature shall be 140°F. The performance tests of 4.6.2 shall be applied.

4.6.3.4 Low temperature. Subject the projector to the low temperature test specified in MIL-STD-810, method 502.1, procedure I. The storage temperature shall be -65°F. The lowest operating temperature shall be -25°F. The perfrmance tests of 4.6.2 shall be applied.

4.6.3.5 Temperature cycling. With power and loads of table I applied, subject the projector to the temperature profile specified in figure 2. At the conclusion of this test, subject the projector to the performance tests of 4.6.2.

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4.6.3.6 Humidity. Place the projector in a humidity chamber and subject it to the humidity test specified in MIL-STD-810, method 507.1, procedure IV. During steps 4, 6 and 8 of procedure IV, subject the projector to the performance tests of 4.6.2. After exposure, subject the projector to the performance tests of 4.6.2.

4.6.3.7 Altitude. Use item 2 of table VI, place the projector in an altitude chamber and subject it to the altitude test specified in MIL-STD-810, method 500.1, procedure I, except that:

a. First, the projector in a non-operating condition, shall be exposed to an atmospheric pressure equivalent to 40000 feet above sea level for a period of not less than 1 hour.

b. Second, with the chamber pressure increased to an atmospheric equivalent to 10000 feet above sea level, subject the projector to the performance tests of 4.6.2.

4.6.3.8 Dust. Subject the projector to the dust test specified in MIL-STD-810, method 510.1, procedure I. At the conclusion of this test, subject the projector to the performance tests of 4.6.2.

4.6.3.9 Salt fog. Subject the projector to the corrosion test specified in MIL-STD-810, method 509.1, procedure I. At the conclusion of this test, subject the projector to the performance tests of 4.6.2.

4.6.3.10 Transient voltages. Subject the projector to transient voltages in accordance with MIL-STD-1275. At the conclusion of this test, subject the projector to the performance tests of 4.6.2.

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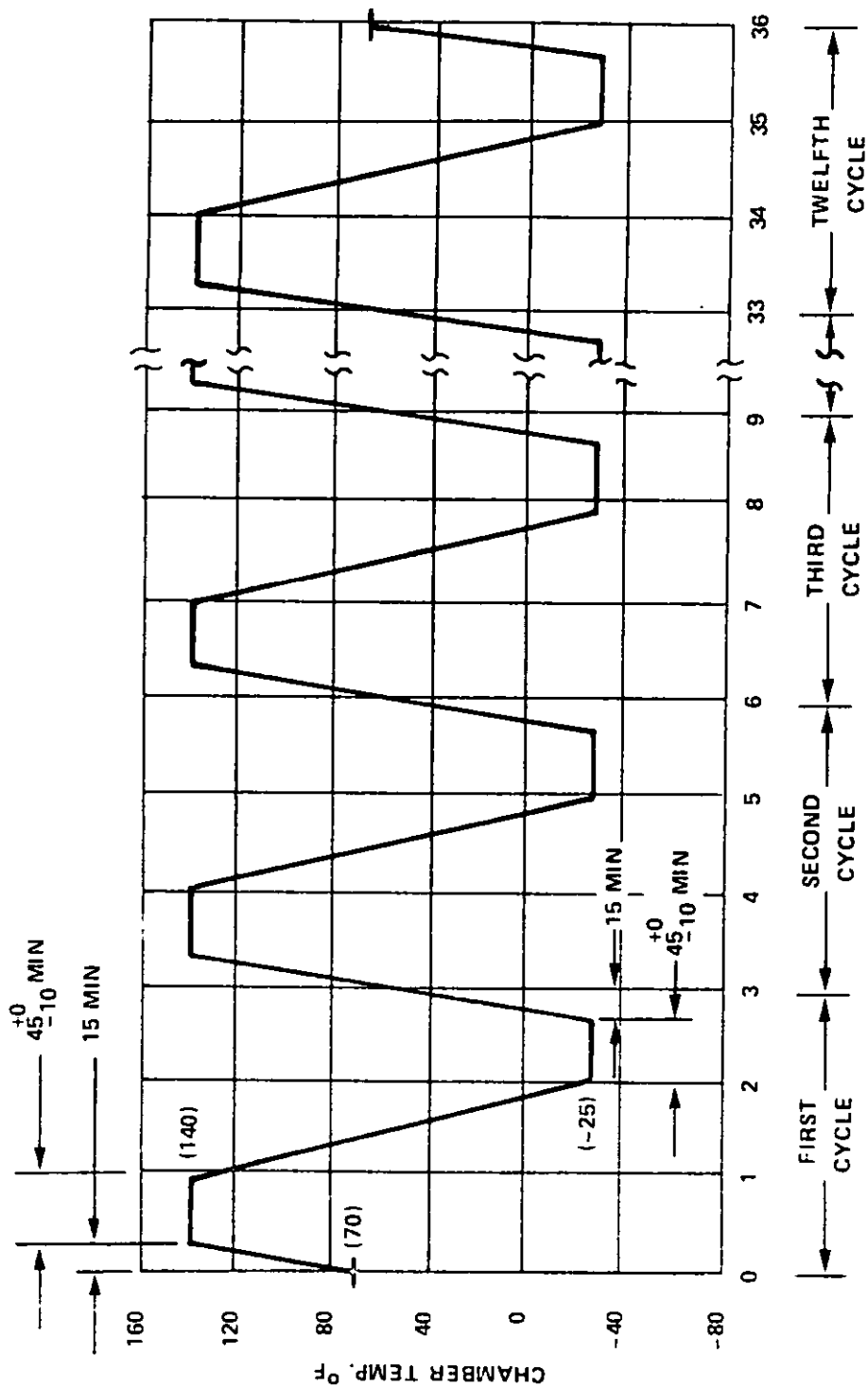


FIGURE 2 Temperature cycling.

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4.6.4 Sealing. Pressurize the projector with dry nitrogen to 5.0 ± 0.1 psig and perform either Sealing A or Sealing B leakage test. After determination of compliance with the sealing requirement, purge the projector with dry nitrogen at 1 psig for not less than 5 minutes and lower the pressure to 0.5 ± 0.1 psig. Nitrogen shall be BB-N-411, type I, class 1, grade B with a dewpoint not greater than -25°F.

4.6.4.1 Test selection. Application of Sealing A and Sealing B shall be as determined by the procurement conditions of 4.5.1 and the first article conditions of 4.4.

4.6.4.2 Sealing A. Use item 7 of table VI and subject the projector to the leakage (immersion) test specified in MIL-STD-810, method 512.1, procedure IV. The immersion period shall be not less than 5 minutes.

4.6.4.3 Sealing B. Use item 7 of table VI, place the projector in the following environment for not less than 2 hours and verify that the pressure drop is not greater than 0.2 psig. During the second hour of the two-hour period, each movable control device on the projector shall be operated three times to its extreme positions.

a.	Temperature	73° ± 2.5°F
b.	Relative humidity	50 percent ± 5 percent
c.	Atmospheric pressure	28.5 + 2.0 -3.0 in. Hg.

5. PACKAGING

5.1 Packaging, packing and marking. The projector shall be packaged, packed and marked in accordance with packaging data sheet P11744800. The level of protection shall be as specified in the procurement document. (See 4.5.6)

6. NOTES

6.1 Intended use. The Reticle Image Projector: M21 is used on tank fire control systems in conjunction with a tank periscope and an electric ballistic computer. Its functions are to project and automaticaly position laterally, a reticle pattern into the periscope's optical system for control of the main gun; to project a reticle pattern into the unity power viewer of the periscope for control of a machine gun.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of preservation, packaging and packing.

c. Applicable packaging data sheet number (see 5.1).

d. Applicable stock number.

e. Provisions for first article testing.

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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DOCUMENT IDENTIFIER (Number) AND TITLE

MIL-P-50775B PROJECTOR, RETICLE IMAGE: 11744800 M21

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

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